

Anticipation stock: pre-building for seasonal peaks, promotions and shutdowns

Part 2 · Categories of stock — solution report with data, method, results and charts

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1. Anticipation stock: pre-building for seasonal peaks, promotions and shutdowns

Annual demand **16,000** units; regular capacity **1,250/month** (+200 overtime); August shutdown runs at **50 %** capacity.

$$E(t-1) = \max(0, D(t) - K(t) + E(t)), \quad E(12) = 0$$

INFEASIBLE — short
1,625 units

Regular capacity only

721 units

Average anticipation stock

Feasible

Regular + overtime

2,050

Overtime units needed

1,775 units (May)

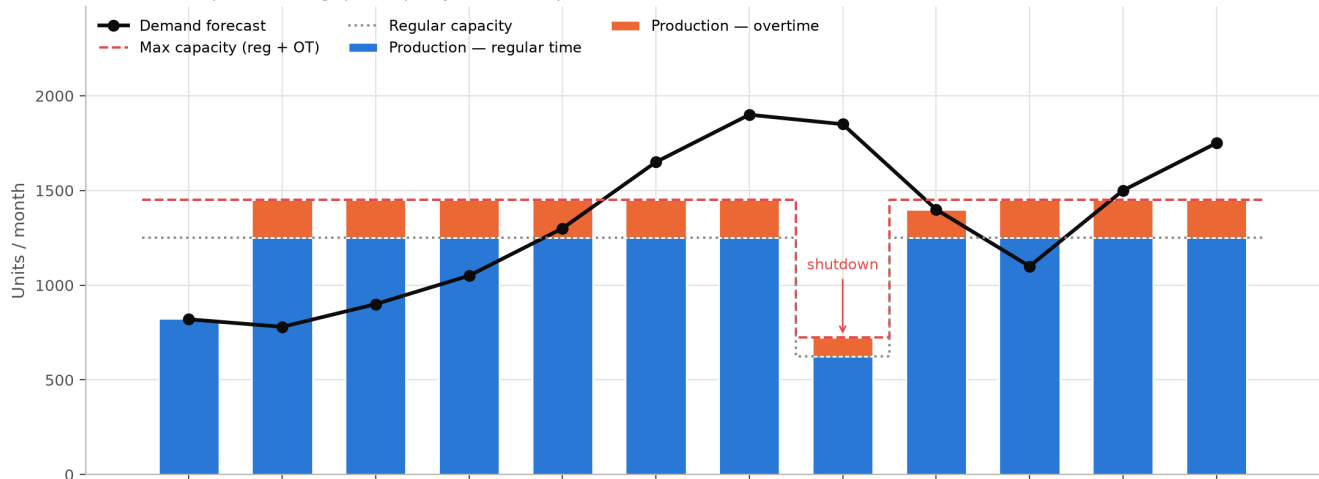
Peak anticipation stock

5

Months demand > capacity

Production plan that minimises anticipation stock

Build as late as possible, using spare capacity before each peak



Anticipation stock carried

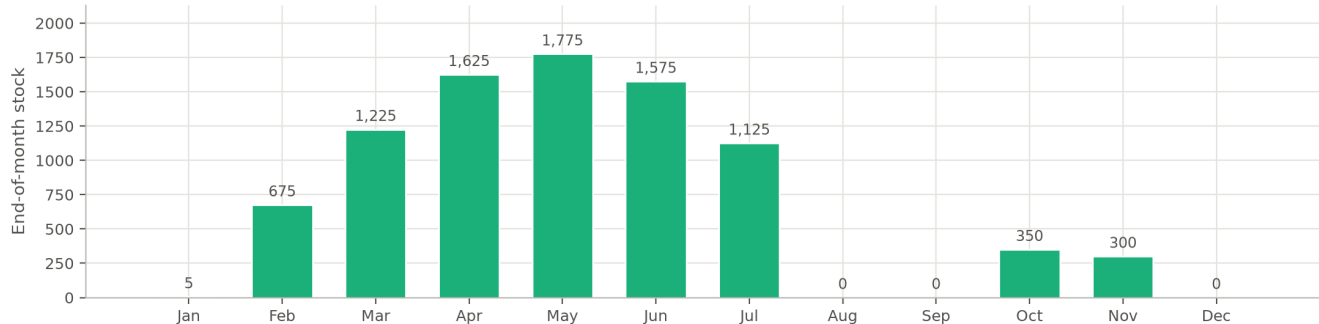


Figure 1.1 — Monthly demand vs capacity, the resulting production plan, and anticipation stock.

Cumulative view: the vertical gap is the anticipation stock

Where cumulative regular capacity falls below cumulative demand, overtime or more stock is required

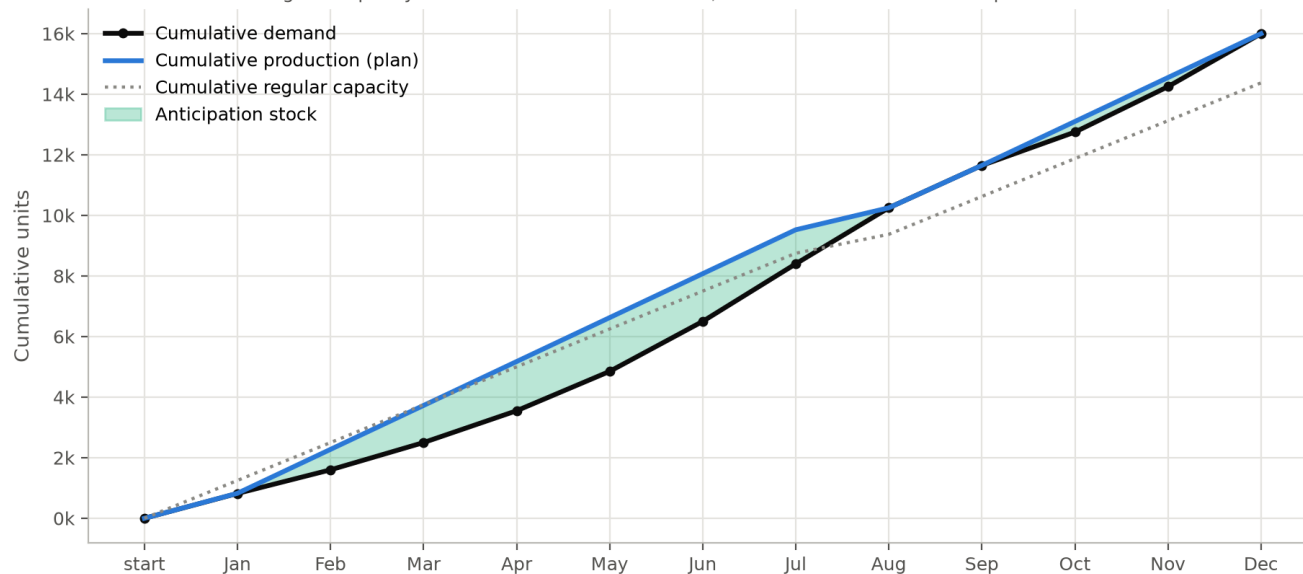


Figure 1.2 — Cumulative demand, production and regular capacity.

Monthly anticipation-stock plan (effective capacities include the shutdown).

month	demand forecast	reg cap eff	total cap eff	regular used	overtime used	production plan	anticipation stock end
Jan	820	1,250	1,450	825	0	825	5
Feb	780	1,250	1,450	1,250	200	1,450	675
Mar	900	1,250	1,450	1,250	200	1,450	1,225
Apr	1,050	1,250	1,450	1,250	200	1,450	1,625
May	1,300	1,250	1,450	1,250	200	1,450	1,775
Jun	1,650	1,250	1,450	1,250	200	1,450	1,575
Jul	1,900	1,250	1,450	1,250	200	1,450	1,125
Aug	1,850	625	725	625	100	725	0
Sep	1,400	1,250	1,450	1,250	150	1,400	0
Oct	1,100	1,250	1,450	1,250	200	1,450	350
Nov	1,500	1,250	1,450	1,250	200	1,450	300
Dec	1,750	1,250	1,450	1,250	200	1,450	0